ECE 3355: Electronics

Section 17184
Fall 2017

Exam 2

Version B - Fall 2017

November 16, 2017

Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. You may use two crib-sheets as described in the syllabus and discussed in class. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: Solution

Question	Points	Score
1	25	
2	30	
3	45	
Total:	100	

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

V_T = thermal voltage = $\dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 6.4 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

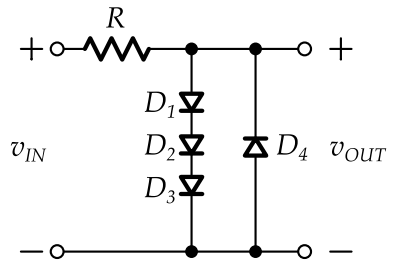
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. For the circuit below, assume $V_{D0} = 0.7$ volts and $r_D = 0$.



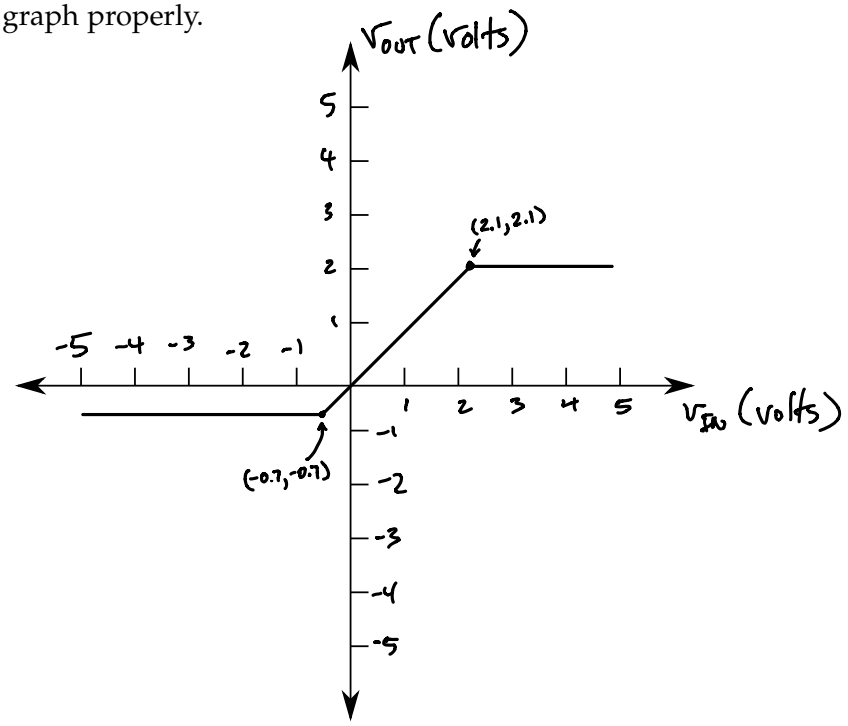
(a) (10 points) Write an expression for v_{OUT} in terms of v_{IN}.

$$v_{OUT} = \begin{cases} 2.1\,V & v_{IN} \geq 2.1\,V \\ v_{IN} & -0.7V < v_{IN} < 2.1\,V \\ -0.7\,V & v_{IN} < -0.7V \end{cases}$$

As v_{IN} increases from 0, all diodes are off (open circuits) and $v_{OUT} = v_{IN}$ until v_{IN} is large enough to forward-bias the diode bank D_1-D_3, which occurs at $v_{IN} = 3 \cdot 0.7V = 2.1\,V$. After that, v_{OUT} remains fixed at 2.1V.

For $v_{IN} < 0$, v_{OUT} tracks v_{IN} until D_4 becomes forward-biased, which happens when $v_{IN} = -0.7V$. After this, the output remains fixed at $-0.7V$

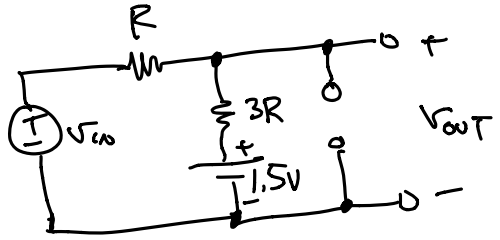
(b) (5 points) Sketch v_{OUT} and v_{IN} for $-4\,\text{volts} < v_{IN} < 4\,\text{volts}$ on the graph provided. Be sure to label the graph properly.



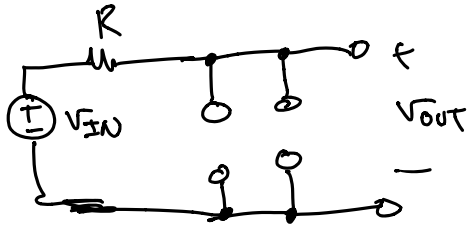
(c) (10 points) Now, use $V_{D0} = 0.5\,\text{volts}$ and $r_D = R$. Write an expression for v_{OUT} in terms of v_{IN}.

$$v_{OUT} = \begin{cases} \frac{3}{4}(v_{IN} - 1.5V) + 1.5V & v_{IN} \geq 1.5V \\ v_{IN} & -0.5V < v_{IN} < 1.5V \\ \frac{1}{2}(v_{IN} + 0.5V) - 0.5V & v_{IN} \leq -0.5V \end{cases}$$

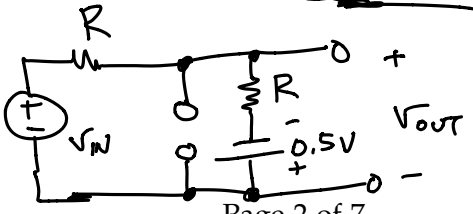
For $v_{IN} \geq 1.5V$



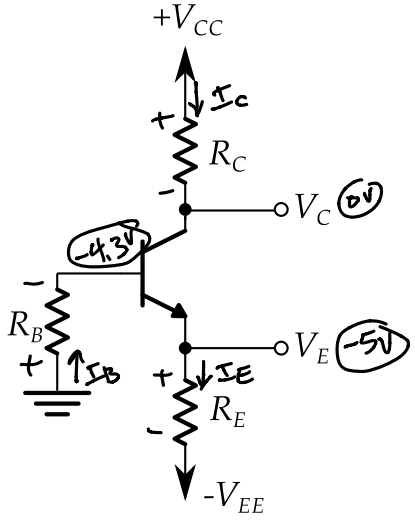
For $-0.5V < v_{IN} < 1.5V$



For $v_{IN} \leq -0.5V$



2. Use the following circuit to complete this problem.



(a) (10 points) Write expressions for V_C, V_E, and the currents in the transistor in terms of R_B, R_C, V_{CC}, $-V_{EE}$, and β.

$$V_{EE} = I_B R_B + V_{BE} + I_E R_E \Rightarrow I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta+1)R_E}$$

$$I_C = \beta I_B$$

$$I_E = (\beta+1) I_B$$

$$V_E = (-V_{EE}) + I_E R_E$$

$$V_C = V_{CC} - I_C R_C$$

(b) (10 points) Find values for R_B, R_C, and R_E so that $V_C = 0$, $V_E = -5\,\text{V}$, and $r_\pi = 250\,\Omega$. Use $V_{CC} = 10\,\text{V}$, $-V_{EE} = -10\,\text{V}$ and $\beta = 100$.

Since $r_\pi = \frac{V_T}{I_B}$, $I_B = \frac{V_T}{r_\pi} = \frac{0.025\text{V}}{250\,\Omega} = 0.1\,\text{mA}$ $\quad I_C = \beta I_B = 10\,\text{mA}$

$V_C = 0\text{V} = V_{CC} - I_C R_C$ so $\boxed{R_C = \frac{V_{CC}}{I_C} = \frac{10\text{V}}{10\text{mA}} = 1\text{k}\Omega}$

$V_E = (-V_{EE}) + I_E R_E$ so $R_E = \frac{V_E - (-V_{EE})}{I_E} = \frac{V_E - (-V_{EE})}{\frac{\beta+1}{\beta} I_C} = \frac{-5-(-10\text{V})}{\frac{101}{100} \cdot 10\text{mA}}$

$\boxed{R_E = \frac{5\text{V}}{\frac{101}{100} \cdot 10\text{mA}} = 495\,\Omega}$

Note that $V_B = -4.3\text{V}$ so $I_B = \frac{4.3\text{V}}{R_B}$ $\quad R_B = \frac{4.3\text{V}}{I_B} = \frac{4.3\text{V}}{0.1\,\text{mA}} = \boxed{43\text{k}\Omega}$

For Version A:

$R_C = 1.5\text{k}\Omega$

$R_E = 495\Omega$

$R_B = 93\text{k}\Omega$

(c) (10 points) What is the range of V_C when $50 \leq \beta \leq 150$?

$$V_C = V_{CC} - I_C R_C = V_{CC} - \beta I_B R_C = V_{CC} - \beta\left[\frac{V_{EE} - V_{BE}}{R_B + (\beta+1)R_E}\right]R_C$$

for $\beta = 50$

$$V_C = 10\text{V} - 50\left[\frac{10\text{V} - 0.7\text{V}}{43\text{k}\Omega + 51\cdot(495\Omega)}\right](1\text{k}\Omega) = 3.2\text{V}$$

for $\beta = 150$

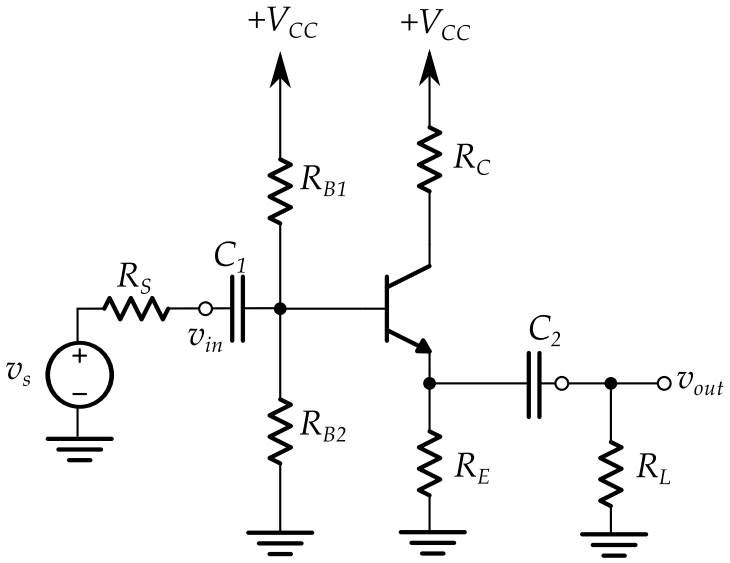
$$V_C = 10\text{V} - 150\left[\frac{10\text{V} - 0.7\text{V}}{43\text{k}\Omega + 151\cdot 495\Omega}\right](1\text{k}\Omega) = -1.85\text{V}$$

$$-1.85\text{V} < V_C < 3.2\text{V}$$

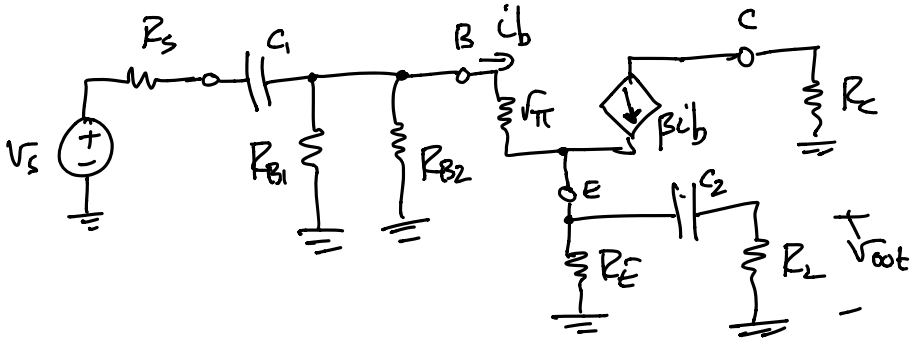
For Version A:

$$-4.18\text{V} < V_C < 5.93\text{V}$$

3. Use the following circuit to complete this problem.

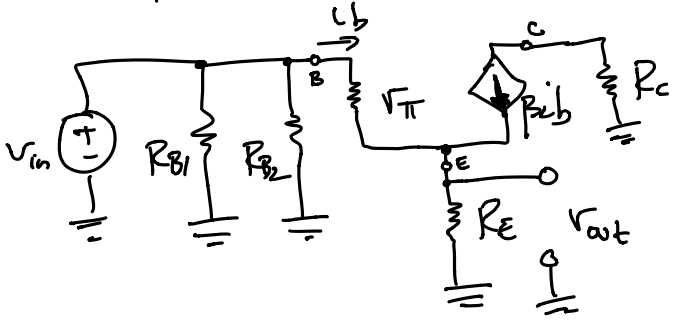


(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

midband: C_1 & C_2 are shorts



$$v_{out} = (\beta+1) i_b R_E$$

$$v_{in} = i_b r_\pi + (\beta+1) i_b R_E$$

$$\frac{v_{out}}{v_{in}} = \frac{(\beta+1) R_E}{r_\pi + (\beta+1) R_E}$$

(c) (10 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + i_b}$$

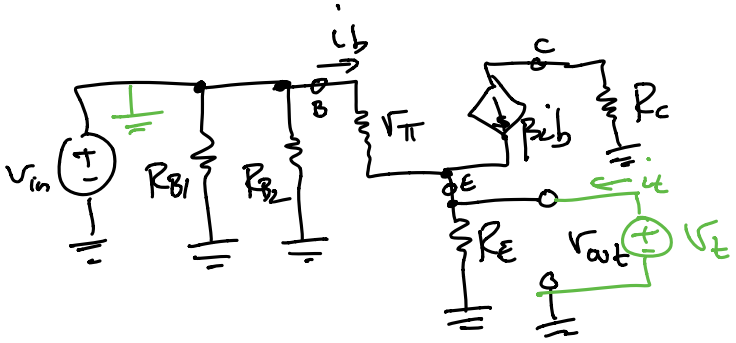
since $i_b = \frac{v_{in}}{r_\pi + (\beta+1)R_E}$

$$R_{in} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + \frac{v_{in}}{r_\pi + (\beta+1)R_E}} = \boxed{R_{B1} \,||\, R_{B2} \,||\, [r_\pi + (\beta+1)R_E]}$$

or

$$R_{in} = R_{B1} \,||\, R_{B2} \,||\, \frac{v_{in}}{i_b} = R_{B1} \,||\, R_{B2} \,||\, \frac{i_b r_\pi + (\beta+1) i_b R_E}{i_b}$$

$$= R_{B1} \,||\, R_{B2} \,||\, [r_\pi + (\beta+1)R_E]$$



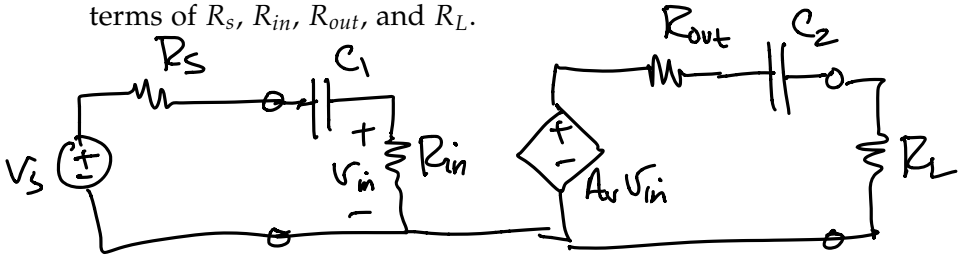
$$R_{out} = \frac{v_t}{i_t} \qquad i_t = \frac{v_t}{R_E} - (\beta+1) i_b$$

$$i_b = -\frac{v_t}{r_\pi}$$

$$R_{out} = \frac{v_t}{\frac{v_t}{R_E} - (\beta+1)\left[-\frac{v_t}{r_\pi}\right]}$$

$$= \frac{1}{\frac{1}{R_E} + \frac{1}{\frac{r_\pi}{\beta+1}}} = \boxed{R_E \,||\, \left[\frac{r_\pi}{\beta+1}\right]}$$

(d) (10 points) Write expressions for the resistances seen by the capacitors C_1 and C_2 in terms of R_s, R_{in}, R_{out}, and R_L.



So, for C_1 : $R_s + R_{in}$

for C_2 : $R_{out} + R_L$

Hence, we have two corners with

$$\omega_1 = \frac{1}{(R_s + R_{in})C_1}$$

$$\omega_2 = \frac{1}{(R_{out} + R_L)C_2}$$

(e) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit.

(label slopes)

